

Decoherence sets a resolution limit, not a bias, in spectroscopic determination of the ANNNI phase diagram

Jeongbin Jo^{1,2,*}

¹*Department of Physics, Yonsei University*

²*School of Mathematics and Computing (Computational Science and Engineering), Yonsei University*

(Dated: September 23, 2026)

We map the phase diagram of the one-dimensional axial next-nearest-neighbour Ising (ANNNI) chain with an optimiser-free eigensolver built on real-time evolution and classical spectral estimation, and ask how depolarizing noise distorts the phase boundaries. It does not distort them in any of the obvious ways. A depolarizing channel applied to the exact ground state rescales every two-point correlator by one common factor, verified to six digits, leaving the order parameters in the same relative order and the fidelity-susceptibility ridge at the same position and width; applied as a decay envelope on the time signal it leaves the estimated energies unbiased, because the estimator reads the phase of its spectral roots and the envelope acts only on their modulus. What survives is resolution. Decoherence and finite sampling together truncate the usable record at $T_{\max} \sim \ln \sqrt{M}/\Gamma$, capping the spectral resolution at $\delta E \sim 2\pi/T_{\max}$. The protocol is a Loschmidt echo on Trotterised evolution, with an explicit depolarizing channel after every two-qubit gate and binomial readout, and Γ is measured at each error rate rather than extrapolated, the ratio Γ/p drifting by a third across the range studied. Over five decoherence rates the resulting criterion is exact in one direction and approximate in the other: no gap below δE is ever recovered, in sixteen attempts, while above it recovery runs at 93% and falls as the record shortens. Since $\Delta \sim |h - h_c|^{z\nu}$ near criticality, the criterion predicts an apparent boundary displacement $|h^* - h_c| \sim \Gamma^{1/z\nu}$ set by the critical exponents; testing that scaling needs a wider usable range in Γ than the estimator currently supports, and we do not claim it here.

I. INTRODUCTION

The one-dimensional ANNNI chain [1] is a standard testbed for quantum algorithms that locate phase transitions, and the question we take up here is posed in an equally standard way: map its phase diagram on a quantum device, then determine how noise distorts the boundaries. The premise built into that question — that noise moves boundaries — deserves checking before it is used, and when we check it for depolarizing noise it largely fails. What follows is an account of which mechanisms do nothing, why, and which one is left.

Two considerations shape the choice of eigensolver. First, a variational loop entangles any observed boundary shift with optimiser convergence, initialisation and ansatz expressivity, which is precisely what one does not want when the shift itself is the measurement. Second, subspace diagonalisation in a Krylov basis needs the complex amplitude and a second observable, which the readout we settle on does not provide; Sec. III A lays that out, and reports a trap we fell into on the way. We therefore use real-time evolution followed by classical spectral estimation, an approach with no optimiser and no generalised eigenproblem, in the family of ODMD and QCELS [3–5].

II. MODEL AND EXACT BASELINE

We study the open chain

$$H = -J \sum_i Z_i Z_{i+1} + J\kappa \sum_i Z_i Z_{i+2} - Jh \sum_i X_i, \quad (1)$$

with $J = 1$ and $\kappa > 0$ frustrating the ferromagnetic nearest-neighbour term. The model has a ferromagnetic phase at small κ and small h , a period-four antiphase for $\kappa > 1/2$ at small h , and a paramagnetic phase at large h , with a multiphase point at $\kappa = 1/2$, $h = 0$. We take as reference boundaries

$$h_c(\kappa) = \frac{1 - \kappa}{\kappa} \left(1 - \sqrt{\frac{1 - 3\kappa + 4\kappa^2}{1 - \kappa}} \right), \quad \kappa < \frac{1}{2}, \quad (2)$$

$$h(\kappa) \simeq 1.05 \sqrt{\kappa - \frac{1}{2}}, \quad \kappa > \frac{1}{2}. \quad (3)$$

Figure 1 is the exact-diagonalisation ground truth against which everything below is measured: a 41×41 grid at $N = 6$, the same size as the circuit runs, showing the structure factors $S(k=0)$ and $S(k=\pi/2)$ that detect ferromagnetic and antiphase order, the transverse magnetisation $\langle X \rangle$, and the fidelity susceptibility $\chi_F = 2(1 - |\langle \psi_h | \psi_{h+\delta} \rangle|)/\delta^2$. The last of these assumes no order parameter and nevertheless traces both analytical boundaries.

* jeongbin033@yonsei.ac.kr

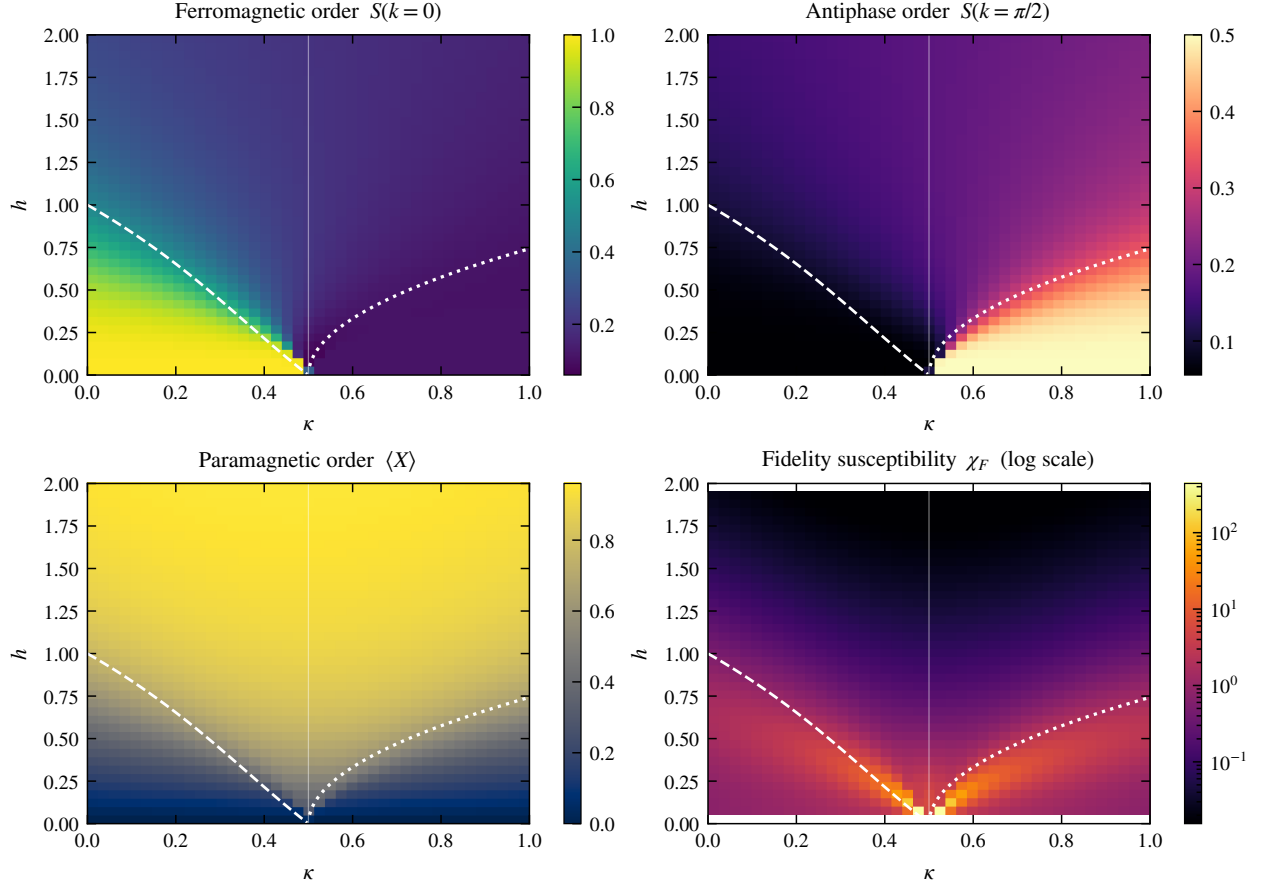


FIG. 1. Exact-diagonalisation phase diagram of the ANNNI chain, $N = 6$, open boundaries, on a 41×41 grid in (κ, h) . Dashed and dotted white curves are the analytical ferromagnet–paramagnet and antiphase–floating boundaries. The model-free fidelity susceptibility (lower right) reproduces both without assuming an order parameter. The $h = 0$ row is excluded from that panel: the ground state is exactly degenerate there, so the overlap between neighbouring states is not defined.

III. OPTIMISER-FREE SPECTRAL ESTIMATOR

We measure the real-time signal

$$g(t) = \langle \psi_0 | e^{-iHt} | \psi_0 \rangle = \sum_k |c_k|^2 e^{-iE_k t} \quad (4)$$

on a uniform grid in t and recover the low-lying spectrum by ESPRIT [2], that is, from the shift-invariance of the Hankel matrix built from g .

A. Why not Krylov

Real-time Krylov consumes the same record. Its overlap and Hamiltonian matrices are Toeplitz in the time index, $S_{jk} = g((k-j)\delta t)$ and $H_{jk} = \langle \psi_0 | H U((k-j)\delta t) | \psi_0 \rangle$, so neither costs the $O(D^2)$ measurements a naive count suggests, and the Toeplitz form holds for a Trotterised propagator as exactly as for an exact one — we checked the Gram matrix against it at 5×10^{-14} .

Two things rule it out here and neither is accuracy. It cannot run on the echo at all: S needs the complex amplitude, and $|g|^2$ has lost the phase, so choosing the readout of Sec. III C forecloses Krylov independently of how well it would have worked. And it needs a second observable, $\langle \psi_0 | H U(t) | \psi_0 \rangle = i dg/dt$, which is either a numerical derivative of a noisy signal or a separate measurement — one circuit per Pauli term, fifteen of them at $N = 6$, against one for g .

Figure 2 maps what it does when it can run, over subspace dimension and Trotter step, with no noise and no shots. Given both matrices from the same propagator (left) it converges and then plateaus, and the plateau is the Trotter bias of the propagator it was handed: 1.1×10^{-2} at $\delta t = 0.40$ falling to 3.6×10^{-5} at $\delta t = 0.10$, as δt^2 . Given both from exact evolution (right) there is no plateau and it reaches machine precision. Neither is fragile.

At equal model dimension ESPRIT is nonetheless ahead. At $\delta t = 0.30$, $D = 24$, Krylov returns a relative error of 1.3×10^{-2} where ESPRIT at order 24 on the same record returns 2.6×10^{-4} ; Krylov needs $D \gtrsim 48$

to match it, and only the exact-matrix variant improves beyond that.

The middle panel is a warning rather than a property of the method, and we include it because we walked into it. Building S from the Trotterised record and H from exact evolution — two propagators, one matrix each — diverges by five orders of magnitude, up to 4×10^5 where the gap is 0.25. The lower panel shows why: $\text{cond}(S)$ climbs from 10^8 to 10^{18} across this range of D , and the mismatched error tracks it while the consistent one stays flat. Conditioning degrades for both pairings; only the one carrying an inconsistency has anything for it to amplify. The failure is invisible at small D , where both pairings agree, and appears only when the subspace is enlarged — which is exactly what one does to improve a variational method. Our first reading of this data was that Krylov is fragile to Trotter error. It is not; we had mixed our matrix elements.

B. Circuit

Figure 3 shows the two readouts we compared and the evolution they share. The first is a Hadamard test: an ancilla is prepared in $|+\rangle$ and controls the evolution of the register,

$$\rho_{\text{anc}} = \frac{1}{2} \begin{pmatrix} 1 & g^*(t) \\ g(t) & 1 \end{pmatrix}, \quad (5)$$

so that $\langle X \rangle_{\text{anc}} = \text{Re } g$ and $\langle Y \rangle_{\text{anc}} = \text{Im } g$. The evolution is a second-order Trotter product: for a term cP the step is $\exp(-icP\delta t)$, realised as IsingZZ($2c\delta t$) for $P = Z_i Z_j$ and $R_X(2c\delta t)$ for $P = X_i$, with $c = -1$ on nearest-neighbour bonds, $c = +\kappa$ on next-nearest bonds and $c = -h$ on the field, the field applied in two half layers around the couplings. We use the fourth-order Suzuki product, five such substeps with the fractal coefficients, and charge depolarizing noise after every two-qubit gate rather than once per step, so that the higher order pays for its five times the gates. At $N = 6$, $\delta t = 0.30$ the fourth-order circuit reproduces exact evolution to 3.1×10^{-4} against 4.7×10^{-2} for second order at the same step; the residual is Trotter error, and $g(0) = 1$ to machine precision.

C. The echo readout

The second readout carries no ancilla. Writing the state preparation as $|\psi_0\rangle = V|0\rangle^{\otimes N}$, the circuit applies $V^\dagger U(t)V$ to $|0\rangle^{\otimes N}$ and counts how often all N qubits return zero. That probability is

$$P(t) = |\langle 0|V^\dagger U(t)V|0\rangle|^2 = |\langle \psi_0|U(t)|\psi_0\rangle|^2 = |g(t)|^2, \quad (6)$$

the Loschmidt echo, or return probability [7]. Equivalently it is the expectation of the projector $\Pi_0 = |\psi_0\rangle\langle\psi_0|$ on the evolved state, which is how we measure it in simulation; on hardware the two are the same experiment.

Two protocols realise it. One evolves forward under H and back under $-H$, measuring how much of the state returns; the other, used here, inverts the state preparation instead, which costs one extra copy of V rather than a second evolution and is the cheaper of the two when V is shallow relative to $U(t)$. Both are standard, and the Loschmidt amplitude has been measured on trapped-ion hardware for finite-energy properties of the Fermi-Hubbard model [8]. That the Fourier transform of the echo exposes the level structure is also established: peaks of $\tilde{P}(\omega)$ sit at the gaps between pairs of eigenvalues, and the ones involving the ground state are the prominent ones [9]. What we add is not the technique but a quantitative account of when it can and cannot locate a phase boundary.

Expanding $|\psi_0\rangle = \sum_k c_k |E_k\rangle$ in eigenstates turns Eq. (6) into

$$\begin{aligned} P(t) &= \sum_{j,k} |c_j|^2 |c_k|^2 e^{-i(E_j - E_k)t} \\ &= \sum_k |c_k|^4 + 2 \sum_{j < k} |c_j|^2 |c_k|^2 \cos[(E_k - E_j)t], \end{aligned} \quad (7)$$

which we have checked against direct evolution to 3×10^{-14} . Three properties follow, and all three matter downstream.

The spectrum holds level *differences*, not levels. There is no absolute energy anywhere in Eq. (7), so $E_1 - E_0$ cannot be identified by ordering the way it can in the Hadamard test; it has to be found among the differences.

Each line carries weight $|c_j|^2 |c_k|^2$, the *product* of two occupations, against $|c_k|^2$ for the Hadamard test. A level the reference barely populates is therefore suppressed quadratically, which is why Sec. III G spends so much effort on the reference state.

The signal does not decay to zero. The $j = k$ terms give a constant $\sum_k |c_k|^4$, and under depolarizing noise the register tends to the maximally mixed state, whose overlap with $|\psi_0\rangle$ is $\text{Tr}(\Pi_0/2^N) = 2^{-N}$. The envelope in Sec. III H is therefore fitted to $|P(t) - 2^{-N}|$, not to $|P(t)|$.

D. Reading a gap out of a difference spectrum

We take the smallest positive frequency ESPRIT returns, excluding a small window around DC. That rule is not obviously correct, because Eq. (7) contains a difference for every pair of populated levels and nothing forbids one of them from lying below $E_1 - E_0$. Indeed some do: at $(\kappa, h) = (0.2, 0.7)$, $N = 6$, the echo spectrum has lines at 0.0046, 0.0303 and 0.0454, all below the gap of 0.3053.

The rule survives because those lines are weak — weights of 10^{-6} to 10^{-4} , against 1.7×10^{-2} for $E_1 - E_0$, since they connect pairs of highly excited states that the reference hardly populates. A subspace method of order M fits only the M strongest modes, so “smallest frequency” in practice means “smallest among the

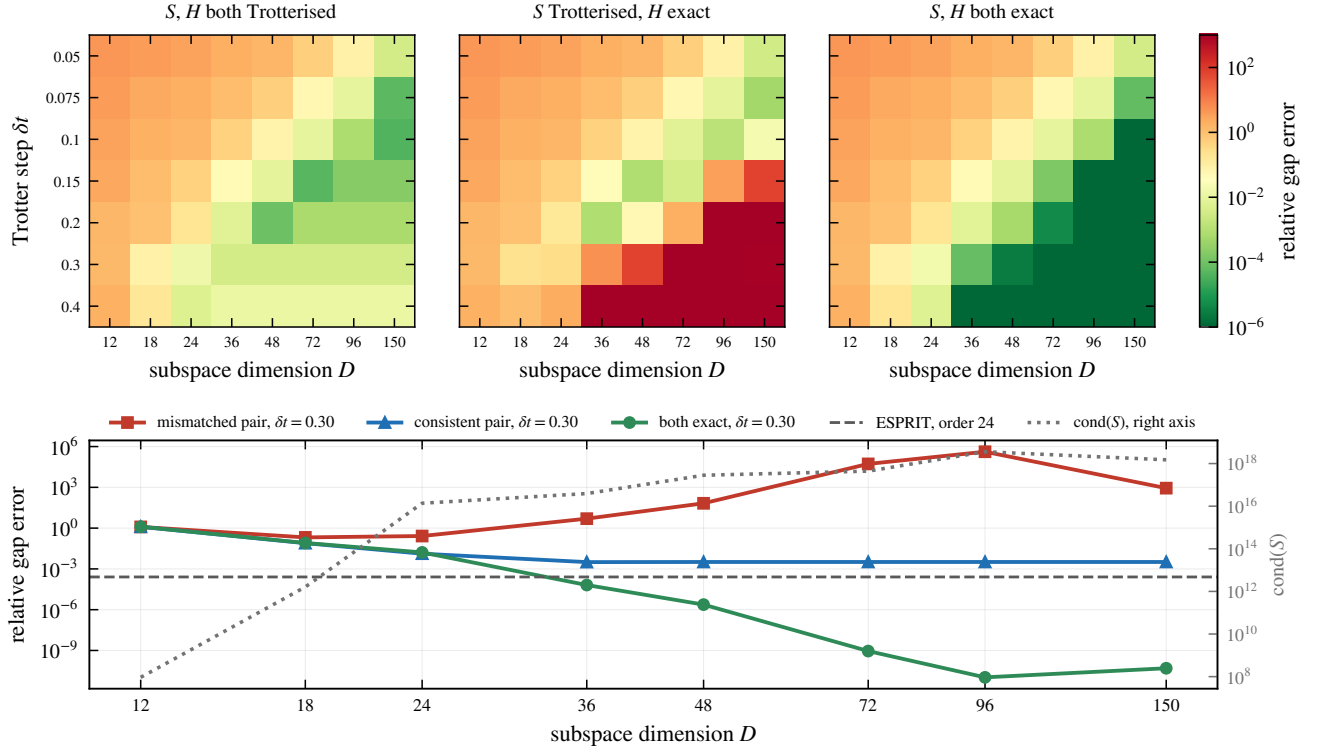


FIG. 2. Relative error of the Krylov gap estimate at $\kappa = 0.2$, $h = 0.66$, $N = 6$, over subspace dimension and Trotter step. No noise and no shots anywhere. Left: both matrices from the Trotterised propagator — convergence to a plateau set by the Trotter bias. Middle: S Trotterised, H from exact evolution — divergence that grows with D . Right: both exact — machine precision. Lower panel: the three pairings at $\delta t = 0.30$ against the condition number of S on the right axis, with ESPRIT on the same record for reference. Below $D \approx 24$ the three are indistinguishable, which is why the failure is easy to miss.

strongest”. Among the sixteen strongest lines at that point, the smallest frequency is exactly $E_1 - E_0$, and the estimator returns 0.3034.

This makes the model order do double duty. It is the usual signal-to-noise trade-off, and it is also what keeps the weak low-frequency clutter out of the fit — which is why raising it does not monotonically help, and why order 48 collapses to 38% in Table III where order 16 reaches 92%.

E. Choosing between the two readouts

Controlling every gate is not cheap. One second-order step costs 9 two-qubit gates uncontrolled and 21 controlled, and the ancilla must couple to all N register qubits, which on hardware means all-to-all connectivity or a chain of swaps. The echo of Eq. (6) needs neither ancilla nor controlled gates. Its disadvantages are the two properties of Eq. (7) noted above — quadratic weights and no ordering — and they are visible in Table I.

They are not, however, decisive, because the ancilla is not free either. It is entangled with the register through every controlled gate, so it carries a decoherence channel of its own, and the two quadratures $\langle X \rangle$ and $\langle Y \rangle$ must share the shot budget. Comparing the two readouts at

TABLE I. The two readouts on the same fields, $N = 6$, reference ref5, each with its own best model order. Left: the noiseless 30-point validation grid. Right: noisy, at matched $\Gamma = 0.104$ and 10^4 shots. Percentages count sites whose exact gap exceeds $2\pi/T$.

readout	noiseless		noisy, matched Γ	
	order	recovered	order	recovered
Hadamard test	32	91%	12	33%
Loschmidt echo	16	87%	24	100%

matched Γ rather than matched gate error — the fair comparison, since Γ is what enters Eq. (9) — the echo recovers every field above the resolution limit where the Hadamard test recovers two of six. We therefore use the echo, and accept that the noiseless map is somewhat worse for it.

The model order differs between the two columns because it must: a subspace method can afford as many modes as the signal-to-noise ratio supports, and no more. Choosing it on clean data and applying it to noisy data was an error we made and had to undo — order 32, optimal noiseless, recovers 12% of the noisy fields.

With depolarizing noise charged per two-qubit gate

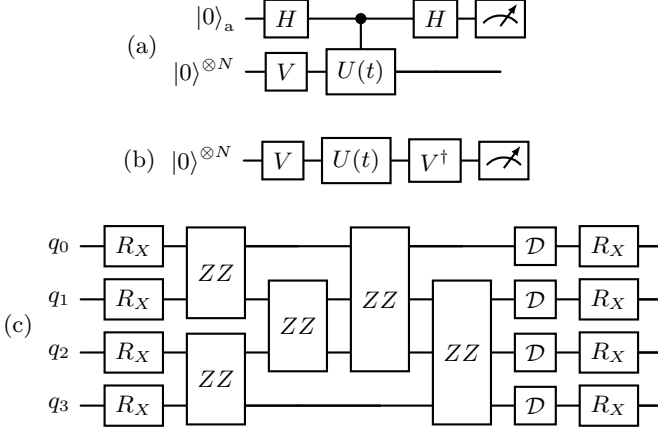


FIG. 3. Two ways to obtain the signal, and the evolution both of them use. (a) Hadamard test: an ancilla controls $U(t)$, and its X and Y expectations give $\text{Re } g$ and $\text{Im } g$. Every gate of U must be controlled, and the ancilla must reach every register qubit. (b) Loschmidt echo: the state preparation is undone and the all-zeros outcome counted, giving $|g|^2$ with no ancilla and no controlled gates. Absolute energies are lost, but the spectrum of $|g|^2$ consists of level differences, which is what the gap detector needs. (c) One second-order step of U , drawn for $N = 4$; the runs use $N = 6$. Reading left to right: half a layer of R_X for the field, the nearest-neighbour ZZ couplings in two non-overlapping columns, the next-nearest couplings in two more, the depolarizing channel $\mathcal{D}$, and the second half of the field. Each ZZ is an IsingZZ rotation; the next-nearest boxes are drawn across three wires but act on the outer two. $\mathcal{D}$ is shown once per step for legibility, but in the simulations it follows every two-qubit gate. Fourth order is five such steps with Suzuki coefficients.

rather than per step, the two readouts and the two Trotter orders can be compared on one footing. Measured this way, the fourth-order Suzuki product at $\delta t = 0.30$ accumulates $\Gamma = 134.7p$ against $67.1p$ for second order at $\delta t = 0.15$, a factor of two that tracks the gate count (45/0.30) versus (9/0.15). Its far smaller Trotter error, 5×10^{-4} against 2×10^{-2} , does not repay that, because Trotter error was never the binding constraint here — Γ is.

Running any of these over the full (κ, h) grid and all decoherence rates is not affordable, so the circuit is used to fix and check an effective model — envelope times sampling noise — which then carries the sweeps. Section III H reports that check.

F. How far the echo scales

The quadratic weighting of Eq. (7) sets a hard ceiling on system size, and it is worth making that explicit because it decided the size of every run reported here. With M shots the return probability is resolved to about $M^{-1/2}$, so the line at $E_1 - E_0$ is visible only while

$$|c_0|^2 |c_1|^2 \gtrsim M^{-1/2}. \quad (8)$$

TABLE II. Weight of the $E_1 - E_0$ line in the echo spectrum at $(\kappa, h) = (0.2, 0.7)$, against the corresponding Hadamard-test line $|c_1|^2$. At $M = 10^4$ the sampling floor is 10^{-2} .

N	$ c_0 ^2 c_1 ^2$ (echo)	$ c_1 ^2$ (Hadamard)	ratio
4	3.8×10^{-2}	7.7×10^{-2}	2.0
6	1.7×10^{-2}	3.7×10^{-2}	2.2
8	1.1×10^{-2}	4.0×10^{-2}	3.6
10	6.3×10^{-3}	3.3×10^{-2}	5.2
12	3.5×10^{-3}	2.4×10^{-2}	6.8

Both occupations shrink as the Hilbert space grows and a fixed reference spreads over more of it, so the product shrinks twice as fast. Table II tracks it for the reference used here.

The echo crosses the floor between $N = 8$ and $N = 10$, and that is where we saw it fail: an earlier map at $N = 10$ recovered 47% of the sites above the resolution limit where the same protocol at $N = 6$ recovers 87%. The Hadamard-test line, being linear in the occupation, is still a factor of five clear at $N = 10$, and the advantage grows with size. The general statement — that the echo is exponentially small in system size and impractical much beyond ten sites — is known, and subsystem variants have been proposed to get around it [10]. At $N = 6$ we are comfortably inside the usable regime; at $N = 10$ we are not, and no amount of estimator tuning recovers that.

G. Reference state

The reference must have weight on the level being tracked, and getting that right across the diagram took two corrections.

First, the spin flip $P = \prod_i X_i$ commutes with H , and $|+\rangle^{\otimes N}$ is an eigenstate of it, so its overlap with the odd-parity partner of the ferromagnetic doublet is exactly zero, measured at 6×10^{-26} . Second, and less obvious, the open chain is also symmetric under reflection R about its centre, and a reference built from $|+\rangle^{\otimes N}$, the ferromagnet $|00 \cdots 0\rangle$ and the palindromic period-four state is *exactly* reflection-even, $|\langle \psi_0 | R | \psi_0 \rangle| = 1.000$. Over much of the plane E_0 is reflection-even while E_1 is reflection-odd, so that reference cannot see E_1 either: at $(\kappa, h) = (0.6, 1.2)$ its overlap is 4×10^{-33} .

Adding two domain-wall components, $|000111\rangle$ and $|001111\rangle$, fixes both. A domain wall is the natural low-lying excitation of the chain and is maximally asymmetric under R , and the resulting five-component reference has worst-case overlap 7.7×10^{-3} on a 6×5 grid against 3.6×10^{-31} for the three-component one.

The improvement is not cosmetic. Table III scans the reference, the ESPRIT model order and the amplitude threshold together on a 30-point validation grid, scoring the fraction of points where $E_1 - E_0$ is recovered to 5%. The three-component reference saturates at 73% of the

TABLE III. Recovery of $E_1 - E_0$ to within 5% on a 30-point noiseless validation grid, echo readout, $N = 6$, $n_t = 800$. *ref3* is the three-component reference, *ref5* adds two domain walls. The right column counts only sites whose gap exceeds $2\pi/T = 0.026$.

reference	order	threshold	all sites	above $2\pi/T$
ref3	16	0.00	63%	73%
ref3	32	0.00	77%	73%
ref5	12	0.00	77%	88%
ref5	16	0.00	80%	92%
ref5	24	0.00	73%	85%
ref5	32	0.00	63%	73%
ref5	48	0.00	33%	38%

TABLE IV. Envelope calibration for the echo circuit, $N = 6$, $\delta t = 0.30$, depolarizing after every two-qubit gate. Γ is fitted to $|P(t) - 2^{-N}|$, which is where the echo decays to, not to zero.

p	Γ	Γ/p	fit residual
1.5×10^{-4}	0.0109	72.5	0.036
3.7×10^{-4}	0.0257	69.4	0.089
7.5×10^{-4}	0.0483	64.5	0.175
1.5×10^{-3}	0.0844	56.3	0.301
3.0×10^{-3}	0.1400	46.7	0.424

sites whose gap exceeds $2\pi/T$; adding the domain walls lifts that to 92%. Order 16 is best here, and the scan falls away on both sides of it — sharply above, where the extra modes fit numerical noise.

H. Validating the effective noise model

The analysis leading to Eq. (9) assumes the channel damps g by $e^{-\Gamma t}$ and that finite sampling puts a floor under it. The second is exact by construction here: the echo returns a probability and we sample it binomially, so the readout noise is the measurement statistics and not a model of them.

The first is an approximation, and measuring it is what fixes Γ . Table IV fits the envelope at each error rate. The decay is close to exponential while the noise is weak — residual 0.036 at $p = 1.5 \times 10^{-4}$ — and departs from it as p grows, reaching 0.42 at $p = 3 \times 10^{-3}$. More importantly the rate is not linear in p : Γ/p falls from 72 to 47 over the same range. Extrapolating a linear fit, as we did at first, puts the record length and therefore δE at the wrong value, so every rate quoted below is measured on the circuit it describes.

We also record a negative result about the detector. The textbook curvature estimator $-d^2E/dh^2$ is unusable at this accuracy: a second derivative on a grid of spacing dh amplifies energy error by $dh^{-2} = 10^4$, so an energy er-

ror of 10^{-3} produces a curvature error of order 30 against a physical curvature of order 11. Under noise levels that changed the energies by less than 0.005, the extracted boundary jumped between 0.42 and 0.77. We use gaps instead, which the estimator returns at no extra cost.

IV. WHAT DEPOLARIZING NOISE DOES NOT DO

A. A channel on the exact ground state

Local depolarizing noise at rate p maps $Z_i \rightarrow (1 - 4p/3)Z_i$, so every two-point function is rescaled by one common factor $\lambda^2 = (1 - 4p/3)^2$. Numerically the ratio $\langle Z_i Z_j \rangle_p / \langle Z_i Z_j \rangle_0$ is constant across all pairs to six digits: 0.947378 at $p = 0.02$ and 0.751111 at $p = 0.10$, against $(1 - 4p/3)^2 = 0.947378$ and 0.751111. Every order parameter shrinks by the same factor, so their relative ordering, and with it the boundary, is untouched. The mixed-state fidelity susceptibility is equally insensitive: its peak sits at $h = 0.525$ with full width 0.250 for $p = 0, 0.03$ and 0.08 alike, only the height falling from 5.82 to 5.24.

B. A decay envelope on the time signal

Depolarizing noise during the evolution multiplies Eq. (4) by $e^{-\Gamma t}$. ESPRIT fits roots $z_k = e^{(-iE_k - \Gamma)t}$ and extracts energies as $\arg(z_k)/-dt$, which is blind to $|z_k|$. A uniform envelope therefore cancels identically, and it does: the mean energy error is 0.0025 at $\Gamma = 0$ against 0.0063 at $\Gamma = 0.40$. With unlimited shots one simply divides the envelope out.

These two null results are what make the remaining mechanism non-obvious, and we regard them as part of the result rather than as discarded attempts.

V. RESOLUTION-LIMITED BOUNDARIES

The envelope becomes consequential only in combination with finite sampling. With M shots per time point the measured $g(t)$ carries noise of size $M^{-1/2}$ per quadrature, so the record is informative only while $e^{-\Gamma t}$ stays above that floor:

$$T_{\max} = \frac{\ln \sqrt{M}}{\Gamma}, \quad \delta E \simeq \frac{2\pi}{T_{\max}} = \frac{2\pi\Gamma}{\ln \sqrt{M}}. \quad (9)$$

Beyond $T_{\max}$ the record is noise, and we truncate there, as an experimenter would. Two levels are then distinguishable only if their spacing exceeds δE , which gives the criterion

$$\Delta(\kappa, h^*) = \delta E(\Gamma, M) \quad (10)$$

for the location h^* at which a boundary first becomes visible.

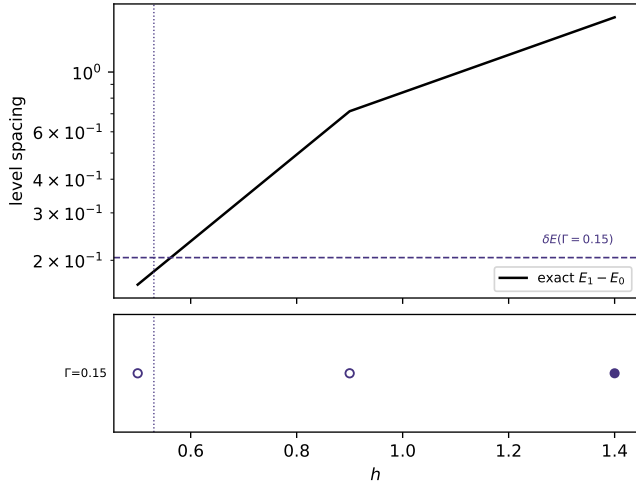


FIG. 4. Resolvability criterion at $\kappa = 0.2$, $N = 6$, 10^4 shots, five decoherence rates. Upper panel: the exact doublet splitting $E_1 - E_0$ (solid) against the resolution limit δE of Eq. (9) for each rate (dashed), with dotted verticals at the predicted onsets where the two cross. Lower panel: what the estimator returned, filled where it recovered the splitting and open where it did not. Every rate switches from open to filled at its own crossing, and no point below δE is ever recovered; above it, recovery degrades as Γ grows and the record shortens.

It matters that this is a statement about noisy records. ESPRIT is a subspace method, not a Fourier transform, and with a clean signal it resolves splittings far below $2\pi/T$: in the noiseless phase map of Sec. VI the Hadamard test recovers half of the sites whose gap lies under that value, though the echo used there does not. What sets the floor in Eq. (9) is not the record length by itself but the record length at which the decayed signal meets the sampling noise. Decoherence does not blur the spectrum; it shortens the part of it that carries information.

Figure 4 tests Eq. (10) at $\kappa = 0.2$, $N = 6$, $M = 10^4$, over five decoherence rates and fifteen fields. Γ is measured at each error rate rather than extrapolated: the ratio Γ/p drifts from 72 to 47 across the range, so a linear fit would put the record length — and with it δE — at the wrong value.

The criterion is exact in one direction and approximate in the other. Below δE the gap is never recovered: sixteen field values across the five rates, without a single exception. Above δE recovery is 93%, 92% and 92% at the three weakest rates, falling to 64% and then 22% as Γ grows. Clearing the resolution limit is therefore necessary but not sufficient, and what fails at strong noise is the estimator rather than the physics: a short record leaves the signal subspace too poorly conditioned for ESPRIT, which is the same ceiling that fixes the model order in Table I. Taken together the criterion predicts the outcome at 61 of 75 points.

Near criticality $\Delta \sim A|h - h_c|^{z\nu}$, so Eqs. (9) and (10)

give

$$|h^* - h_c| \sim \left(\frac{2\pi\Gamma}{A \ln \sqrt{M}} \right)^{1/z\nu}. \quad (11)$$

The apparent boundary is displaced by an amount controlled by the critical exponents, and only logarithmically improvable by spending more shots. For the Ising universality class of the ferromagnet–paramagnet line, $z = \nu = 1$ and the displacement is linear in Γ .

VI. PHASE MAP FROM THE CIRCUIT

Figure 5 is the same measurement run over a 21×21 grid in (κ, h) at $N = 6$, noiseless, with the settings fixed in Table III and not touched afterwards. Each site costs one echo record of $n_t = 800$ points, about 2.4 s of simulation, and the whole map took eighteen minutes; the same map through the Hadamard test took two and a half hours.

The gap is recovered to 5% at 308 of the 441 sites, and at 87% of those whose exact gap exceeds $2\pi/T = 0.026$. Below that value the echo recovers nothing, where the Hadamard test on the same grid recovers about half: super-resolution survives only when the spectrum is sparse enough to pick a weak line out of, and the echo, carrying every pairwise difference, is not. The failures are concentrated in the bottom rows, $h \lesssim 0.25$, where the doublet splitting drops under 10^{-4} and the estimator returns the next distinct difference instead.

Set against the exact map on the right, the circuit reproduces the V-shaped valley that opens at the multiphase point $\kappa = 1/2$ and the gradient that tracks both analytic boundaries. What it does not reproduce is the deep ordered phase, and that is the honest content of the figure: a spectroscopic phase map is limited by what the reference state populates and by how small a splitting the record can support, not by the physics of the model.

VII. DISCUSSION

The practical content of Eq. (11) is an inversion of the usual reading. Decoherence enters this measurement in the same structural role that finite size plays in a classical simulation: as a resolution cutoff that displaces an apparent critical point by a power of the cutoff set by the exponents. That suggests using Γ deliberately as a scaling axis — collapsing $h^*(\Gamma)$ at several κ onto one scaling function and reading the exponents off noisy data — rather than treating noise purely as something to mitigate. Whether that collapse holds quantitatively, and whether the effective exponent is the Ising one or is modified, is the natural test and is the main item left open here.

Two limitations bound the present results. The scaling law Eq. (11) is verified only through its underlying criterion Eq. (10), at five decoherence rates but a single κ ; the

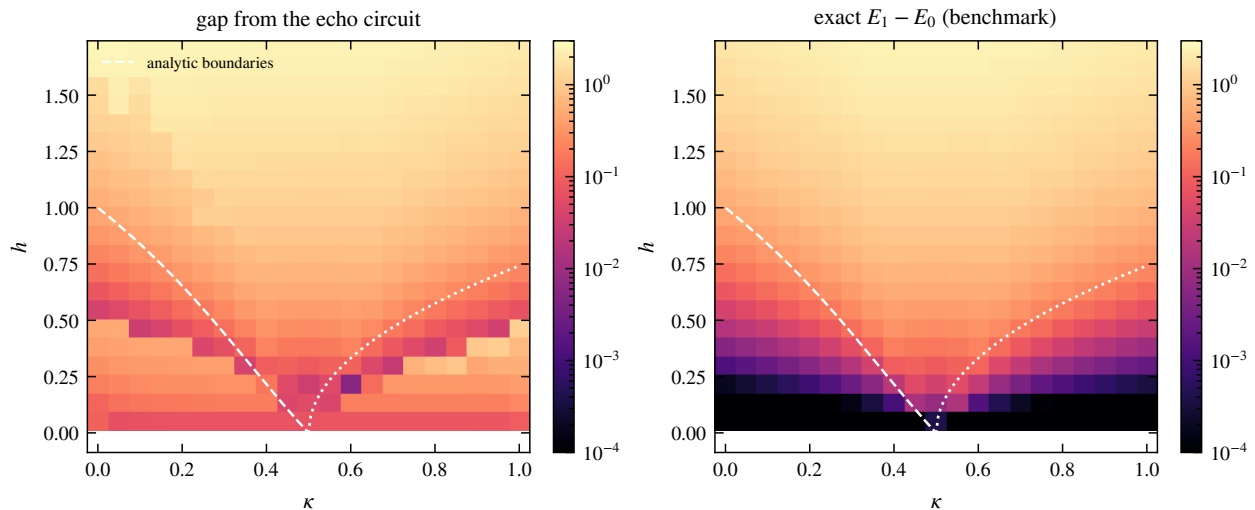


FIG. 5. Gap map from the echo circuit (left) against exact diagonalisation (right), $N = 6$, 21×21 grid, noiseless, $n_t = 800$. Dashed and dotted white curves are the analytic ferromagnet-paramagnet and antiphase-floating boundaries. The circuit tracks the exact map wherever the doublet splitting is large enough to be populated and resolved; the bright band at the bottom is where it is not, and the estimator reports a higher level instead.

collapse itself needs several κ and a wider usable window in Γ than we have. Recovery above δE is already down to 64% at $\Gamma = 0.084$ and 22% at $\Gamma = 0.14$, so the window is roughly $\Gamma \lesssim 0.05$, which is too narrow to fit a power law through. Widening it is a signal-processing problem, not a circuit one: it calls for denoising the Hankel subspace, by Cadzow iteration or the Fourier-denoised variants of ODMD [4], and not for a larger model order, which we tried and which made matters worse. Sec. III F bounds the other axis — more qubits would sharpen the physics but push the echo line under the sampling floor. Second, the uniform depolarizing channel studied here is in a specific sense the benign case: it acts only on the modulus of the spectral roots, which is why it produces no bias. Correlated or non-Markovian noise corrupts their phase and should bias the energies directly, and we expect that, not depolarization, to be where noise genuinely moves a boundary rather than merely hiding it.

VIII. CONCLUSION

We set out to measure how depolarizing noise distorts the phase boundaries of the ANNNI chain and found that, in the two most natural readings of the question, it does not distort them at all. A channel applied to the exact ground state rescales every correlator by one factor and moves nothing; a decay envelope on the time signal leaves a phase-reading estimator unbiased. What noise does is shorten the record that carries information,

and the boundary is then locatable only where the gap clears the resulting resolution $\delta E \simeq 2\pi\Gamma/\ln\sqrt{M}$. That criterion is exact as a necessary condition — across five decoherence rates, not one gap below δE was ever recovered — and holds at better than 90% as a sufficient one while the noise is weak.

Three things had to be right before any of that could be measured, and each was wrong at first. The reference state must break both the spin-flip and the reflection symmetry of the chain, or the level being tracked is invisible at any resolution. The model order of the subspace estimator must be chosen on data matched to the noise it will meet, since the noiseless optimum recovers 12% of noisy fields. And the decoherence rate must be measured on the circuit rather than extrapolated, because Γ/p is not constant. We report these because they are the substance of making such a measurement work, and because each of them, left uncorrected, produces a plot that looks plausible and is wrong.

CODE AVAILABILITY

The code behind every figure and table in this paper, together with a notebook that runs the circuit study end to end, is available at <https://github.com/dolf3131/ANNNI-research>. The computed data is not distributed with it: everything reported here regenerates from the scripts, and shipping the caches would only let a stale file pass for a fresh result.

[1] W. Selke, *The ANNNI model — theoretical analysis and experimental application*, Phys. Rep. **170**, 213 (1988).

[2] R. Roy and T. Kailath, *ESPRIT — estimation of signal parameters via rotational invariance techniques*, IEEE

- Trans. Acoust. Speech Signal Process. **37**, 984 (1989).
- [3] Y. Shen *et al.*, Observable Dynamic Mode Decomposition, <https://github.com/QuantumComputingLab/odmd>.
 - [4] *From noisy observables to accurate ground-state energies: a quantum-classical signal subspace approach with denoising*, APL Quantum **2**, 046103 (2025), [arXiv:2505.10735](https://arxiv.org/abs/2505.10735).
 - [5] *Estimating eigenenergies from quantum dynamics: a unified noise-resilient measurement-driven approach*, Quantum **9**, 1836 (2025), [arXiv:2306.01858](https://arxiv.org/abs/2306.01858).
 - [6] *Ground state energy and magnetization curve of a frustrated magnetic system from real-time evolution on a digital quantum processor*, [arXiv:2401.03015](https://arxiv.org/abs/2401.03015).
 - [7] A. Goussev, R. A. Jalabert, H. M. Pastawski and D. Wisniacki, *Loschmidt echo*, Scholarpedia **7**, 11687 (2012).
 - [8] *Measuring the Loschmidt amplitude for finite-energy properties of the Fermi-Hubbard model on an ion-trap quantum computer*, PRX Quantum **5**, 030323 (2024).
 - [9] *Excited-state quantum phase transition and Loschmidt echo spectra in a spinor Bose-Einstein condensate*, [arXiv:2211.15142](https://arxiv.org/abs/2211.15142).
 - [10] *Probing quantum many-body dynamics using subsystem Loschmidt echos*, [arXiv:2501.16995](https://arxiv.org/abs/2501.16995).
 - [11] *Efficient measurement-driven eigenenergy estimation with classical shadows*, [arXiv:2409.13691](https://arxiv.org/abs/2409.13691).